

## Examiner reports

### Q1.

Nearly all students made good progress with part (a), with over half scoring full marks. Nearly all students successfully found the maximum possible friction and made a comparison with the applied horizontal force. The most frequently seen error was when students wrote a statement that was incorrect, often along the lines of “friction is greater than 150N so there is equilibrium”. This showed a misunderstanding of the friction model and could not be overlooked.

Part (b) was also well answered, but a lack of detail often meant that students had not fully justified their answer. Several impressive and detailed answers were seen, where students explained clearly that the horizontal component of the applied force was greater than the maximum friction so that the crate would not remain stationary.

Resolving of forces was generally successful.

### Q3.

While this question was generally done well there were more errors than in the first two questions. In part (a) the most common error was to include a force directed down the slope. Part (b) was done well, but a few students gave the answer 29.4 and a few used  $\sin 40^\circ$  instead of  $\cos 40^\circ$ . Part (c) was done very well and follow through marks were awarded to those with errors arising from part (b). Part (d) was a little more demanding, with the most common error being to work with only one force. There were many good answers to part (e), but some students gave answers that were not about the forces. For example, a statement like “The box is a particle.” did not relate to the forces acting on the box as was requested in the question.

### Q4.

This question was generally done well. There were very many good force diagrams, with the most common error probably being the omission of arrow heads. Part (b) was done very well. While there were many good answers to part (b) some candidates simply calculated  $4 \times 3 = 12$ . There were also a few candidates who made a sign error, producing equations such as  $F - 50 = 12$ . Candidates were generally able to apply  $F = \mu R$  to their answers from the previous parts, and many gained the follow through marks that were available. There were many good explanations for part (c), but also a number of confused answers, in which it was claimed that the coefficient of friction would have to increase.

### Q5.

The candidates made very little progress with this question and did not seem to be able to deal with the resolving required.

### Q6.

Part (a) was done well by some candidates, but there were a significant number who proceeded as if this was a two particles connected by a string over a peg type of problem. Many times a pair of equations such as  $18g - T = 18a$  and  $T - 12g = 12a$  were seen. This made it impossible for candidates to complete this part of the question. In part (b)(i), some candidates were able to get back on track, as they had the correct

equation for the 18 kg particle, but some did use the acceleration from part (a) instead of  $3 \text{ ms}^{-2}$ . In (b)(ii), there were many correct responses, but a few gave the answer 36 N, suggesting that there was some confusion between “reaction force” and “resultant force”. The responses to part (b)(iii) were quite mixed with some good solutions. It was not uncommon for candidates to simply assume that the friction was equal to 36 N. Other candidates tried to find the difference between a “tension” which was often incorrect and the 36 N.

There were a number of good answers to part (c), but some were too vague. For example just writing “particles” was not enough as it did not make the connection that it was the block that was being modelled as a particle.

### Q7.

Part (a) was done successfully by the vast majority of the candidates. The force diagrams in part (b) were usually good, but the most common errors were to omit either the normal reaction or friction forces. Also some candidates used lines rather than arrows to represent the force.

In part (c), the candidates were often able to produce the printed answer, but in some cases did not show enough working to justify the award of full marks. For questions like this, candidates should be advised to show an answer to more than three significant figures before giving their final answer.

Those who started part (d) by resolving to find the friction force generally did well and completed this part of the question. A number of candidates did make significant errors by assuming, for example, that the friction force was  $2g$  or in some cases  $4g$ .

### Q8.

Parts (a), (b) and (c) were completed well by the majority of candidates and most of the errors were to be found in part (d). In part (a), some candidates made errors such as omitting arrowheads or using inappropriate labels, such as 4 kg instead of  $4g$ . In part (d), although there were many correct answers, the candidates produced a wide range of incorrect answers. Common errors included candidates’ dividing either the 30 N (from the question) or 11.76 N (the friction) by the mass; other candidates calculated the resultant force and not the acceleration.

### Q9.

In part (a) many candidates lost marks for not including arrow heads or due to poor labelling of the forces. Several candidates did not show the two tensions as being equal on their diagrams. There were quite a few good attempts at part (b), with attempts at resolving vertically. Part (c) proved to be very challenging, although there were a number of good solutions. A number of candidates gained an M1 mark for the use of the friction law. The most common errors were to omit one force in their equation or to not include the ‘ $ma$ ’ term in their equation of motion.

### Q10.

This question was found to be quite difficult by many of the candidates. In part (a) there were a number of incorrect force diagrams, including errors such as including a force acting directly down the slope. In parts (b) and (c) there were a lot of confused answers, mainly due to incorrect attempts at resolving. For example, in part (b), one error was to

resolve the normal reaction into components instead of the weight. However, many candidates gained full marks for part (d), where follow-through marks were awarded when the candidates' values from parts (b) and (c) were used correctly in the friction law to find a coefficient of friction.

### Q11.

This question was also done well by many candidates. There were relatively few errors on the force diagrams. In part (b)(ii), some candidates made errors, the most common being to multiply the 30 by 0.5 or subtract the 30 from 49. Part (c) caused difficulties for some candidates. The most common errors here were to attempt to include the weight or to omit the friction force.

### Q12.

There were many correct force diagrams on the scripts, but some candidates lost marks due to missing arrows or poor labelling. The printed answer in part (b) seemed to help the candidates as there were many correct answers. It was clear that in some cases candidates had needed to change their approach to obtain the correct value. There were also many good answers to parts (c) and (d), but some candidates included extra forces which meant that their friction value was wrong. A few candidates seemed to confuse the coefficient of friction and the friction force.

### Q13.

The candidates found part (a) of this question relatively straightforward, but part (b) was found to be very demanding, and very few correct solutions were seen. Candidates lost marks in part (a) mainly due to errors in the resolving: for example using  $\sin 30^\circ$  instead of  $\cos 30^\circ$  in part (a)(ii). A few candidates lost one mark because they gave their final answer for the coefficient of friction as 0.31 rather than 0.313.

In part (b), very few candidates realised that they needed to express the normal reaction in terms of the tension. This error resulted in solutions that were flawed. Most of the candidates who did express the normal reaction in terms of the tension went on to gain full marks for this part of the question, although a few did make arithmetic or algebraic errors.

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### Q14.

Most candidates did well on this question, having few difficulties finding the magnitude of the friction force. In part (b), the great majority of candidates took a correct approach to finding the acceleration. There were a few candidates who did not consider the block and the particle as two separate bodies. Part (c) was done well with candidates experiencing very few difficulties.

### Q15.

Generally part (a) was done well, whilst part (b) caused more difficulties. The fact that the acceleration was given did seem to help many candidates, but some did not justify the minus sign. Applying the constant acceleration equation to find the distance was also done well, although a few candidates took  $u = 0$  and  $v = 4$ . There were a number of candidates who used two constant acceleration equations, finding the time taken to stop as an intermediate step. A few of these candidates did not go on to find the distance.

In part (b), many of the candidates were able to find the magnitude of the friction force. In

a few cases the candidates did not show how the reaction force was obtained. Full marks were not awarded if the candidates simply stated  $R = 1.84$  without any justification. When calculating the acceleration there were many sign errors in candidates' equations and some candidates also included a "4" which appeared as if it was a force. While many candidates realised that the puck would remain at rest, only a relatively small number of them were able to give a convincing explanation. Many of the explanations contained vague statements and lacked precision and clarity.

### Q16.

Part (a) was done very well and some candidates had clearly been helped by the presence of the printed answer. There were several examples of candidates who obtained the printed answer in part (b) showing enough supporting working. A value of 4000 was often introduced without any working to show that it had been obtained from an application of Newton's Second Law.

Several candidates found the tension in the rope first and then found the value of  $P$ . Some candidates were confused about when to include the tension in the rope. The responses to part (c) were variable, but those who had taken a systematic approach to part (b) usually did well. The answers for part (d) were very varied, with some very good explanations and other responses which lacked coherence.

### Q17.

The candidates made good attempts at this question. Some found it difficult to get going in part (a). Most of the candidates who had difficulties had tried to consider particle  $A$  instead of particle  $B$ . Generally the candidates made good attempts at parts (b) and (c). In part (d), the main problems were for candidates who found forming an equation of motion difficult. Often, these candidates had also found part (a) difficult. Most candidates were able to make an attempt at part (e) using their answers to parts (c) and (d).

### Q18.

The force diagram in part (a) was often drawn well, but the usual errors of extra forces and missing labels or arrows were seen. Part (b) was done very well, with the printed answer helping almost all of the candidates to gain full marks. The responses to part (c) were very varied, with some solutions excellent and some very poor. One fairly common error was to form an equation with a sign error, like  $F - 5g \sin 40^\circ = 5 \times 0.8$ . Other more serious errors included using the normal reaction force instead of the weight component parallel to the slope, and simply assuming that the friction was equal to 4. Part (d) of the question was found to be very difficult, with very few candidates giving acceptable answers. The majority of candidates stated that the coefficient of friction would increase.

### Q19.

Almost all candidates gained some marks on this question, but only a fairly small proportion gained full marks. In part (a), there were many good answers, but a number of candidates stated assumptions that were not about the string. Instead they produced comments about the block or the peg. Part (b) was generally done quite well, but there were some candidates who did not gain credit because they did not show enough working. For example, simply stating  $6/4 = 1.5$  was not sufficient. There were, in contrast, some very good solutions that fully justified the final answer.

In part (c), the candidates who considered the forces on the 7 kg sphere did quite well.

The main error that these candidates made was to use  $T - 7g = 7a$ , rather than  $7g - T = 7a$  and obtained a tension of 79.1 N. Other candidates produced equations that involved the 13 kg block but could make little satisfactory progress because they did not know the friction force.

Part (d) was the most demanding part of the question. It required the candidates to find the friction force and then to use  $F = \mu R$ . Some candidates worked through these stages, but there were relatively few complete solutions. The most common errors were to assume that the friction was equal to the tension and simply apply  $F = \mu R$ , and to try to work with a combination of both bodies instead of just the block.

### Q20.

Although there were many correct force diagrams, there were also a significant number which included friction or other forces as well as the two required. A good number of candidates were able to find the acceleration when the slope was smooth, although there were a few candidates who did not show enough working, simply giving statements like “ $a$

$= \frac{1}{2}g = 4.9 \text{ m s}^{-2}$ ”. In part (b), many candidates were able to find the acceleration successfully, but there were two common errors. The first was to assume that they could use  $u = 5 \text{ v}$  without any justification. The second was to simply divide the distance by the time. When finding the friction force, some candidates used  $F = ma$  to obtain a friction force of 7.5 N. There were also a number of correct solutions. When finding the coefficient of friction, many candidates with incorrect values for the friction benefited from the follow through marks and gained full marks. There were however some cases where the normal reaction was incorrect either because no resolving had taken place or because the resolving had been incorrect.

In part (c), there were some good answers from candidates, but there were also quite a few which stated that the acceleration would be less, but did not give a good enough explanation for this.

### Q21.

A surprising number of diagrams in part (a) failed to include all the forces. Part (b) was very successful, showing a good understanding of the friction law. Part (c) was done well by many, but others couldn't appreciate the significance of the constant speed, the necessary starting point of the explanation. Many were successful in part (d), although some confused the normal reaction with the vertical component of the tension. Part (e) was generally done well with candidates gaining credit for correct calculations using their components of the tension force.

### Q22.

Parts (a)(i), (ii), (v) and (b) proved routinely successful for well prepared and accurate candidates. Parts (a)(iii) and (iv) tested the understanding of the forces in the static situations and proved highly discriminating.

### Q23.

Diagrams in part (a) were clear and mostly correct, with occasional marks lost due to missing arrows or an additional or incorrect force; the reaction force being vertical and the



weight being given as 'g' were the most frequent errors. Part (b) was mostly done correctly, but part (c) was less successful, with extra forces sometimes present in equilibrium equations. The friction law was usually applied in part (d) but marks were lost through missing or incorrect inequality signs and rounding of answers to 2 significant figures.

#### Q24.

Diagrams in part (a) were clear and mostly correct, with occasional marks lost due to missing arrows or a missing or incorrect force, with the weight being given as 'g' as the most frequent. Parts (b) and parts (c) were also good, but few scored well in part (d), with one or even two forces missing from the equation of motion, the weight component being the most frequent omission.

#### Q25.

Again diagrams were mostly good, and parts (a)(ii) and (c) usually gained full marks. Part (b) was successfully completed by most candidates with the main error being the inclusion of the weight of P instead of the frictional force in the equation of motion of P. Pleasingly very few candidates used a 'whole string' approach instead of forming equations of motion for the two particles as requested. In part (d) the intended response focussed on the remaining force acting on each particle and the subsequent change in speed. Many candidates were successful here but others focussed on forces no longer present.

#### Q26.

There was a good number of correct force diagrams. Some candidates added one or both of the components of the tension to their force diagram. In some cases these appeared as if they were additional forces. Parts (b) and (c) were done well by most candidates who seemed to have benefitted from the printed answers. There were some candidates who did not gain full marks, because they did not show sufficient working. In part (d), many candidates obtained the correct value. Some candidates used the two printed answers as expected, if they had not been able to obtain the two values for themselves. There were some very good answers to part (e), but there were also quite a few confused ones as well, with many candidates finding it hard to give a good answer.

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#### Q27.

Candidates found part (a) quite straightforward, although some did omit g from their calculations. Part (b) was done well by some candidates, but some of the weaker candidates used the weight of the 8kg particle rather than the tension when forming an equation to find its acceleration.

#### Q28.

There was a good number of correct force diagrams. Some candidates added one or both of the components of the tension to their force diagram. In some cases these appeared as if they were additional forces. Parts (b) and (c) were done well by most candidates who seemed to have benefitted from the printed answers as expected. There were some candidates who did not gain full marks, because they did not show sufficient working. In part (d), many candidates obtained the correct value. Some candidates used the two printed answers as expected, if they had not been able to obtain the two values for themselves. There were some very good answers to part (e), but there were also quite a few confused ones as well, with many candidates finding it hard to give a good answer.

**Q29.**

A variety of force diagrams were seen on the candidates' scripts. Some were very good, but errors included the omission of forces and showing the friction force acting up the slope. In part (b) the candidates were clearly helped by the printed answer and there were many correct responses. In some cases the candidates did not show enough working to gain full marks. Part (c) proved to be very challenging, with only a very small number of candidates getting the correct coefficient of friction. The main problem was that the candidates did not include all of the terms that were needed in their equations of motion.

**Q30.**

This was an unusual question and produced many pleasing responses, showing good understanding of the law of static friction. Part (a) was generally done very well. Part (b), as expected, proved more discriminating, with a significant proportion stating 2.8 N as their answer, while some gave negative frictional forces. Part (c) produced many good solutions.

**Q31.**

This question produced a variety of responses. Parts (a)(i) and (ii) were mostly answered well, although some showed all the forces acting rather than those on particle A as requested. Part (a)(iii) produced irrelevant work, as many failed to appreciate the equilibrium situation, and produced equations of motion as if a dynamic situation existed. Part (a)(iv) proved testing, but many showed a good understanding of the forces involved. Part (b) was generally done well and accurately.

**Q32.**

In part (a) diagrams were often disappointing, with common errors including an additional force in the direction of motion and the inclusion of acceleration or velocity as a force. Parts (b) and (c) were done quite well, but a significant number of candidates selected the wrong component when finding the required forces and some failed to give their final answer to the required degree of accuracy stated in the rubric.

**Q33.**

This question was done quite well, although some did not find the magnitude of the appropriate quantity in part (a), while others conveniently changed signs in the midst of working to obtain a positive answer.

The simplicity of the request in part (b) was not always realised, and in part (c), some could not find the normal reaction force correctly.

**Q34.**

A surprising number found the static part of this question very demanding, with relatively few correct responses to parts (a)(i), (iii) and (iv). Part (b) proved more successful.

**Q35.**

Part (a) proved to be highly discriminating, with few fully correct solutions seen. Candidates showed little understanding of the equilibrium situation of the particles. There was confusion as to which forces acted on each particle, and also between horizontal and

vertical forces. The law of friction was known and used by some candidates, but it was rare to see competent handling of inequalities.

Part (b) was often started well, although the description of the surface as 'smooth' was not always appreciated by candidates. Some of the algebraic work was unconvincing, but candidates who were able to write down correct equations for the particles usually completed their solution correctly.



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